
GENERAL RELATIVITY

Tutorial problem set 14, 19.12.2020, with solutions

■ PROBLEM 36 Tolman-Oppenheimer-Volkoff equation.

In this problem you will examine more closely the general relativistic models of spherical massive objects (such as stars). Namely, you will study spherically symmetric static solutions of the inhomogeneous Einstein equation.

If a star has a radius larger than its Schwarzschild radius, the space-time outside of it will be described by Schwarzschild metric (Birkhoff's theorem). In order to say something about the space-time inside the star we have to solve the Einstein equation for the particular source,

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G_N T_{\mu\nu} , \quad (36.1)$$

meaning we have to know something about the composition of the star.

Since we are looking for a spherically symmetric and static solution, we assume the following line element

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2d\Omega^2 . \quad (36.2)$$

(The relevant geometric quantities for this metric are given at the end of this problem, just specialize to time-independent α and β .)

- (a) Calculate the Ricci scalar, and the non-vanishing components of the Einstein tensor.

The star will be modeled by a perfect fluid, whose energy-momentum tensor is

$$T_{\mu\nu} = (\rho + \mathcal{P})u_\mu u_\nu + \mathcal{P}g_{\mu\nu} , \quad (36.3)$$

where we assume to be in the rest frame of the fluid, $u_\mu = (-e^\alpha, 0, 0, 0)$, and because of the assumed spherical symmetry, $\rho = \rho(r)$ and $\mathcal{P} = \mathcal{P}(r)$.

- (b) Calculate all the non-vanishing components of the fluid's energy-momentum tensor.
- (c) Write down all the independent components of the Einstein equation (there are three of them).
- (d) Notice that the (tt) equation involves only β and ρ . Show that it can be rewritten as

$$\partial_r \left(r - r e^{-2\beta} \right) = 8\pi G_N r^2 \rho . \quad (36.4)$$

Introduce a new function,

$$m(r) = \frac{1}{2G_N} \left(r - r e^{-2\beta} \right) , \quad (36.5)$$

and show that the line element can then be written as

$$ds^2 = -e^{2\alpha}dt^2 + \left(1 - \frac{2G_N m(r)}{r} \right)^{-1} dr^2 + r^2 d\Omega^2 . \quad (36.6)$$

We have substituted function $\beta(r)$ for function $m(r)$ in a way so g_{rr} would look like a Schwarzschild one, but with a 'radially dependent mass'. Because of Birkhoff's theorem, we know that outside of the star the solution must be a Schwarzschild one, and therefore $m(R_*) = M$ must correspond to the total mass of the star (as seen by the distant observer), where R_* is the radius of the star. Find the expression for $m(r)$ in terms of $\rho(r)$, assuming $m(0) = 0$.

- (e) Show that the (rr) equation can be written as

$$\frac{d\alpha}{dr} = \frac{G_N m(r) + 4\pi G_N r^3 p}{r(r - 2G_N m(r))} . \quad (36.7)$$

- (f) Combine the equation from (e) together with the $\vartheta\vartheta$ equation to get

$$\frac{d\mathcal{P}}{dr} = - \frac{(\rho + \mathcal{P})[G_N m(r) + 4\pi G r^3 \mathcal{P}]}{r[r - 2G_N m(r)]} . \quad (36.8)$$

This is the Tolman-Oppenheimer-Volkoff equation for hydrostatic equilibrium. (An easier way of obtaining this equation is to consider the conservation equation for the energy-momentum tensor instead of the $(\vartheta\vartheta)$ equation).

- (g) In order to get a closed system of equations, we need one more apart from the Tolman-Oppenheimer-Volkoff equation, and that is the equation of state for the fluid, $\mathcal{P} = \mathcal{P}(\rho)$ (or some equivalent one). Then, in principle (36.8) represents an equation for $\mathcal{P}(r)$.

One can get a quite simple and semi-realistic model of a star by assuming it is made out of an incompressible ideal fluid. It means that its energy density $\rho(r)$ is constant throughout,

$$\rho(r) = \begin{cases} \rho_* & , \quad r < R_* \\ 0 & , \quad r > R_* \end{cases} \quad (36.9)$$

where R_* is the radius of the star. This assumption serves instead of the equation of state. Solve the Tolman-Oppenheimer-Volkoff equation to find the pressure inside the star. What is the boundary condition for pressure you are going to assume? Express the final answer in the form $\mathcal{P}(r) = \rho_* P(r)$, where P should be expressed in terms of G_N , M , R_* and r .

- (h) By considering the pressure at the center of the star show that the maximum mass a star of the type considered here of a given radius R_* can be

$$M_{max} = \frac{4R_*}{9G_N} , \quad (36.10)$$

i.e. that there are no static solutions for a mass larger than this. This is a special case of the Buchdahl's theorem (but the result holds more generally). Can you give a physical interpretation of this result?

The non-vanishing components of the Ricci tensor for metric (36.2) are

$$R_{tt} = e^{2(\alpha-\beta)} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \alpha \right] , \quad (36.11)$$

$$R_{rr} = - \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta - \frac{2}{r} \partial_r \beta \right] , \quad (36.12)$$

$$R_{\vartheta\vartheta} = e^{-2\beta} \left[r(\partial_r \beta - \partial_r \alpha) - 1 \right] + 1 , \quad R_{\varphi\varphi} = R_{\vartheta\vartheta} \sin^2 \vartheta . \quad (36.13)$$

SOLUTION: (a) The nonvanishing components of the Einstein tensor are given in Eqs. (5.135) of Carroll,

$$\begin{aligned} G_t^t &= -\frac{1}{r^2}e^{-2\beta} [2r\beta' - 1 + e^{2\beta}] \\ G_r^r &= \frac{1}{r^2}e^{-2\beta} [2r\alpha' + 1 - e^{2\beta}] \\ G_\vartheta^\vartheta &= G_\varphi^\varphi = e^{-2\beta} \left[\alpha'' + \alpha'^2 - \alpha'\beta' + \frac{1}{r}(\alpha' - \beta') \right], \end{aligned} \quad (36.14)$$

where $d\Omega^2 = d\vartheta^2 + \sin^2(\vartheta)d\varphi^2$. To get the Ricci tensor, notice that the trace $\text{tr}[G_{\mu\nu}] = G = -R$, such that $R_{\mu\nu} = G_{\mu\nu} - \frac{1}{2}g_{\mu\nu}G$ results in (36.13).

(b) In the fluid rest frame, $u^\mu = (e^\alpha, 0, 0, 0)$, $u_\mu = (-e^{-\alpha}, 0, 0, 0)$, such that the nonvanishing elements of the energy-momentum tensor are,

$$T_t^t = -\rho, \quad T_r^r = \mathcal{P} = T_\vartheta^\vartheta = T_\varphi^\varphi. \quad (36.15)$$

where ρ and $\mathcal{P}$ denote energy density and pressure, respectively.

(c) The independent components of the Einstein's equations are (because of the rotational symmetry, the $(\vartheta\vartheta)$ and $(\varphi\varphi)$ equations are not independent),

$$-\frac{1}{r^2}e^{-2\beta} [2r\beta' - 1 + e^{2\beta}] = -8\pi G_N \rho \quad (36.16)$$

$$\frac{1}{r^2}e^{-2\beta} [2r\alpha' + 1 - e^{2\beta}] = 8\pi G_N \mathcal{P} \quad (36.17)$$

$$e^{-2\beta} \left[\alpha'' + \alpha'^2 - \alpha'\beta' + \frac{1}{r}(\alpha' - \beta') \right] = 8\pi G_N \mathcal{P}, \quad (36.18)$$

(d) Upon multiplying (36.16) by $-r^2$ one obtains,

$$\partial_r (r - r e^{-2\beta}) = 8\pi G_N r^2 \rho, \quad (36.19)$$

which is identical to Eq. (36.4).

Next, from (36.5) it follows that,

$$e^{2\beta} = \frac{1}{1 - \frac{2G_N m(r)}{r}}, \quad (36.20)$$

where $m(r)$ is known as a semilocal mass. Inserting this into (36.2) gives (36.6).

(e) Upon inserting (36.20) into (36.17) and multiplying it by $r^2 e^{2\beta}$ one obtains,

$$\frac{d\alpha}{dr} = \frac{1}{1 - \frac{2G_N m(r)}{r}} - 1 + \frac{8\pi G_N r^2 \mathcal{P}}{1 - \frac{2G_N m(r)}{r}} = \frac{G_N m(r) + 4\pi G_N r^3 \mathcal{P}}{r [r - 2G_N m(r)]}, \quad (36.21)$$

which agrees with (36.7).

(f) First note that the (tt) equation (36.19) implies,

$$m' = 4\pi r^2 \rho(r) \implies m(r) = 4\pi \int_0^r d\tilde{r} \tilde{r}^2 \rho(\tilde{r}) = \int_{\tilde{r} < r} d^3 \tilde{r} \rho(\tilde{r}), \quad (36.22)$$

such that the semilocal mass at a star's radius R , $m(R) \equiv M$ is the mass of the star, with R being its radius.

Secondly, subtracting the (tt) (36.16) and (rr) (36.17) equations yields,

$$\alpha' - \beta' = \frac{2G_N m(r) + 4\pi G_N r^3 (\mathcal{P} - \rho)}{r(r - 2G_N m(r))}. \quad (36.23)$$

Next, taking a derivative of α' in Eq. (36.21) gives,

$$\alpha'' = \frac{4\pi G_N r [\rho + 3\mathcal{P} + r\mathcal{P}']}{r - 2G_N m(r)} - 2 \frac{[G_N m(r) + 4\pi G_N r^3 \mathcal{P}] [r - G_N m(r) - 4\pi G_N r^3 \rho]}{r^2 [r - 2G_N m(r)]^2}, \quad (36.24)$$

where we made use of $m' = 4\pi r^2 \rho$. Multiplying Eq. (36.18) by $e^{2\beta}$ one obtains,

$$\begin{aligned} \alpha'' + \left(\alpha' + \frac{1}{r}\right) (\alpha' - \beta') &= \frac{4\pi G_N r^2 \mathcal{P}'}{r - 2G_N m(r)} + \frac{8\pi G_N r \mathcal{P}}{r - 2G_N m(r)} + \frac{4\pi G_N r [\rho + \mathcal{P}] [G_N m(r) + 4\pi G_N r^3 \mathcal{P}]}{[r - 2G_N m(r)]^2} \\ &= \frac{8\pi G_N r \mathcal{P}}{r - 2G_N m(r)}, \end{aligned} \quad (36.25)$$

where we made use of (36.24), (36.21) and (36.23). Now, the term on the right hand side cancels with the corresponding one at the left hand side, and the remaining terms combine into,

$$\mathcal{P}' = -\frac{(\rho + \mathcal{P}) [G_N m(r) + 4\pi G_N r^3 \mathcal{P}]}{r [r - 2G_N m(r)]} = -(\rho + \mathcal{P}) \alpha', \quad (36.26)$$

which is the Tolman-Oppenheimer-Volkoff equation (36.8) for hydrostatic equilibrium. Show that this equation can be also derived by making use of the covariant conservation of the energy-momentum tensor.

(g) For an incompressible fluid (36.9), and writing $\mathcal{P} = \rho_* P(r)$, Eq. (36.8) simplifies to,

$$P' = -(1 + P)(1 + 3P) \frac{\frac{4\pi G_N \rho_*}{3} r}{1 - \frac{8\pi G_N \rho_*}{3} r^2}, \quad (r < R_*) \quad (36.27)$$

where we made use of $m(r) = \frac{4\pi \rho_*}{3} r^3 = M(r/R_*)^3$ for $r < r_*$ and $m(r) = M$ when $r > R_*$. From Eq. (36.27) it is clear that $r_c = \sqrt{3/(8\pi G_N \rho_*)} = \sqrt{R_*^3/(2G_N M)}$ defines a critical radius which must be greater than the star's radius, *i.e.* $r_c > R_*$, from which one gets, $R_* > 2GM$, which states that star's radius must be greater than the Schwarzschild radius (otherwise there would be no solution for hydrostatic equilibrium and the star would necessarily collapse). Eq. (36.27) can be solved by separation of variables,

$$\begin{aligned} \frac{dP}{(1 + P)(1 + 3P)} &= -\frac{\frac{4\pi G_N \rho_*}{3} r dr}{1 - \frac{8\pi G_N \rho_*}{3} r^2} \\ \Rightarrow \frac{1}{2} \ln \left(\frac{1 + 3P}{1 + P} \right) &= \frac{1}{4} \ln \left(1 - \frac{8\pi G_N \rho_*}{3} r^2 \right) + \text{const.} \end{aligned} \quad (36.28)$$

where the constant ought to be chosen such that the pressure vanishes at the surface of the star $r = R_*$ (in realistic situations, this pressure is not zero, but instead is equal to the atmospheric pressure of gasses at the surface of the star, which we shall neglect in this problem). Choosing $P(R_*) = 0$ determines the integration constant in (36.28), and one obtains,

$$\frac{1 + 3P}{1 + P} = \sqrt{\frac{1 - \frac{8\pi G_N \rho_*}{3} r^2}{1 - \frac{8\pi G_N \rho_*}{3} R_*^2}} = \sqrt{\frac{1 - \frac{2G_N M}{R_*^3} r^2}{1 - \frac{2G_N M}{R_*^3}}} \Rightarrow \frac{\mathcal{P}(r)}{\rho_*} = P(r) = \frac{\sqrt{1 - \frac{2G_N M}{R_*^3} r^2} - \sqrt{1 - \frac{2G_N M}{R_*^3}}}{3\sqrt{1 - \frac{2G_N M}{R_*^3}} - \sqrt{1 - \frac{2G_N M}{R_*^3}}} \quad (36.29)$$

(h) The pressure at the center of the star is,

$$\mathcal{P}(0) = \rho_* = \frac{1 - \sqrt{1 - \frac{2G_N M}{R_*}}}{3\sqrt{1 - \frac{2G_N M}{R_*}} - 1}. \quad (36.30)$$

This quantity cannot be negative and must be less than infinity, which leads to,

$$M < M_{\text{cr}} = \frac{4R_*}{9G_N} \quad (36.31)$$

which is known as Buchdahl's theorem (this should not be considered as a proof of the theorem, but rather as an example in which it is satisfied). To get a physical interpretation of the result, note that Eq. (36.31) can be recast as,

$$R_* > \frac{9}{8}(2MG_N) = \frac{9}{8}r_S, \quad (36.32)$$

stating that the radius of the star in units of the Schwarzschild radius must be greater than $9/8$. Had one demanded that $\mathcal{P}(0) \leq \rho_*$ (which is what one obtains if kinetic term of a field dominates the pressure), one would obtain, $R_*/r_S \geq 4/3$, and had one demanded that $\mathcal{P}(0) \leq \rho_*/3$, which is the relativistic particle fluid limit, one would obtain, $R_*/r_S \geq 9/5$.

Physically, the fact that there is no static solution if (36.31) is not obeyed, means that the star must collapse into a black hole.